

Predicting Gene Expression with Abductive Learning (ABL)

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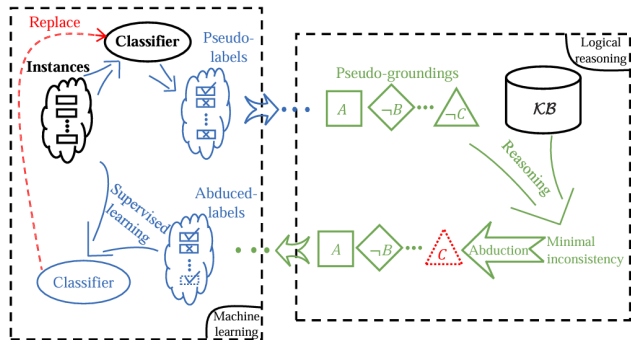
Problems



Abductive Learning (ABL)

Bridging machine learning and logic reasoning

- Learning from data
- Reasoning on structured knowledge
- Iteration between ML & KR
- Semisupervision (Use of unlabeled data)



Formalisation of ABL

- Machine Learning: $y = f(x)$
- Logical consistency: $B \cup x \cup y \models z$
- Objective: ML function f such that:

$$\begin{aligned} &f: x_l \rightarrow y_l \\ \text{s.t. } &B \cup x_u \cup \hat{y} \models z, \text{ or} \\ &B \cup x_u \cup \bar{y} \models z, f \leftarrow \text{train}(f, x_u, \bar{y}) \end{aligned}$$

Background knowledge B , labeled data (x_l, y_l) ,

Unlabeled input x_u , pseudo-label $\hat{y} = f(x_u)$ and revised label $\bar{y}$, observation z .



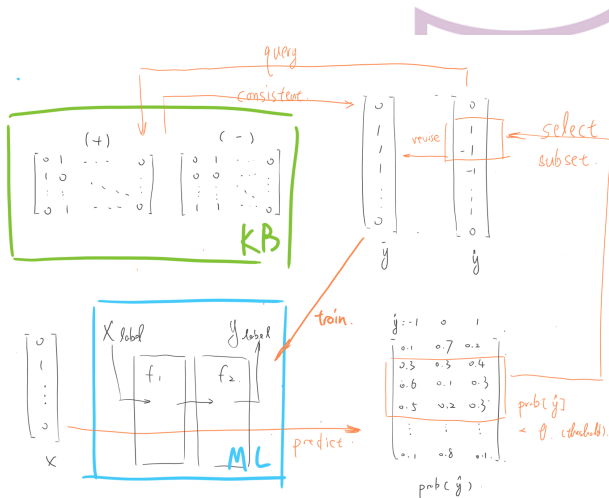
Dataset

- ~ 100 data samples of e.coli MG1655
- Collected from [NCBI SRA](#)
- **Single gene overexpression** over 16 regulators
- Input: one-hot vector
- Output: differential expressed genes over genome (0,1,-1)

	b0002	b0003	b0004	b0031	b0032	b0033	b0048	b0071	b0072
EX01273485	0	0	0	0	0	0	0	0	0
EX01273487	0	0	0	0	0	0	0	0	0
EX01273488	0	0	0	0	0	0	0	0	0
EX01273489	0	0	0	0	0	0	0	0	0
EX01273490	0	0	0	0	0	0	0	0	0
EX01273491	1	1	1	1	1	1	1	1	1
EX01273492	1	1	1	1	1	1	1	1	1
EX01273493	1	0	1	0	1	1	1	1	1
EX01273494	1	1	1	0	0	0	1	0	0
EX01273495	1	0	1	0	0	1	1	1	0
EX01273496	0	0	0	0	0	0	0	0	0
EX01273497	0	0	0	0	0	0	0	0	0
EX01273498	0	0	0	0	0	0	0	0	0
EX01273499	0	0	0	0	0	0	0	0	0
EX01273500	0	0	0	0	0	0	0	0	0
EX01273501	0	0	0	0	0	0	0	0	0
EX01273502	0	0	0	0	0	0	0	0	0
EX01273503	0	0	0	0	0	0	0	0	0
EX01273504	0	0	0	0	0	0	0	0	0
EX01273505	0	0	0	0	0	0	0	0	0
EX01273506	1	1	1	1	1	1	1	1	1
EX01273507	1	1	1	1	1	1	1	1	1
EX01273508	1	1	1	0	1	1	1	0	0
EX01273509	0	0	1	0	1	1	0	0	0
EX01273510	1	1	1	0	1	1	1	0	1
EX02780400	0	0	0	0	0	0	0	0	0
EX02780490	0	0	0	0	0	0	0	0	0
EX02780491	0	0	0	0	0	0	0	0	0
EX02780492	0	0	0	0	0	0	0	0	0
EX02780493	0	0	0	0	0	0	0	0	0
EX02874132	0	-1	0	0	0	0	0	0	1
EX02874133	0	-1	0	0	0	0	-1	0	0
EX02874134	1	-1	0	1	-1	0	-1	1	1
EX02874135	1	-1	0	0	1	1	-1	1	1
EX02874136	1	-1	0	0	1	1	-1	0	1
EX02874137	1	-1	0	0	1	0	-1	0	1
EX02874138	-1	-1	0	0	0	0	-1	0	0
EX02874139	1	0	0	0	1	0	0	1	1
EX02874140	0	0	0	0	0	0	0	0	0
EX02874141	0	0	0	0	0	0	0	0	0
EX02874142	-1	-1	0	0	0	0	-1	0	0
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EX02874144	0	-1	0	0	1	0	0	1	1
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EX02874146	0	-1	0	0	0	0	0	0	0
EX02874147	0	0	0	0	0	0	0	0	1
EX03803290	0	0	0	0	0	0	0	0	1
EX03803291	0	0	1	0	0	0	-1	0	1
EX03803292	0	0	1	0	0	0	0	0	1
EX03803293	-1	0	-1	-1	-1	-1	-1	-1	-1
EX03803294	0	0	1	0	1	1	-1	0	-1
EX03803295	1	1	1	-1	1	1	0	1	1
EX03803296	1	0	1	-1	0	0	-1	0	1
EX03803297	1	1	1	1	1	1	0	1	1
EX03803298	1	1	1	1	1	1	1	1	1
EX03803299	0	0	0	-1	0	0	-1	0	0
EX03803300	0	0	-1	-1	-1	-1	-1	0	0
EX03803301	-1	0	-1	-1	-1	-1	-1	0	-1
EX03803302	1	1	0	0	1	1	-1	1	1
EX03803303	-1	-1	-1	-1	-1	-1	-1	-1	-1
EX03803304	0	0	0	0	-1	-1	-1	0	0
EX03803305	-1	-1	-1	-1	-1	-1	-1	-1	-1

Model Framework

- Learning: semi-supersive
- Reasoning: minimal inconsistency
- Iteration of learning & reasoning



Problems

- Need more data for training and evaluation
- Dual regulations in TRN
- Data - knowledge inconsistency

